MIL-C-31/2A
1 February 1971
SUPERSEDING
MIL-C-81/2
29 January 1968

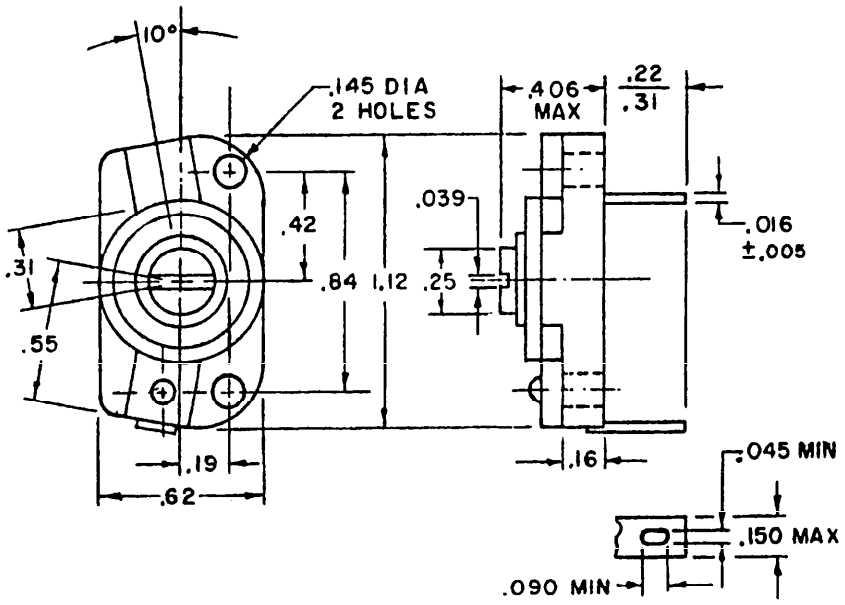
MILITARY SPECIFICATION SHEET

CAPACITORS, VARIABLE, CERAMIC, DIELECTRIC,

STYLE CV14

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-81.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



INCHES	MM
.005	.13
.016	.41
.039	.99
.045	1.14
.090	2.29
.145	3.68
.150	3.81
.16	4.06
.19	4.83
.22	5.59
.25	6.35
.31	7.87
.406	10.31
.42	10.67
.55	13.97
.62	15.57
.84	21.34
1.12	28.45

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm
3. Unless otherwise specified, tolerances are ±.03 (.76 mm), ±.010 (.25 mm) and 5° on angles.

Ⓐ denotes changes.

FSC 5910

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REQUIREMENTS:

Dimensions: See figure.

Capacitance value: See table I.

DC rated voltage: See table I.

Operating temperature range:
-55° to +85° C.

Characteristic: See table I.

Dielectric withstanding voltage:
Method 301 of MIL-STD-202:
A dc potential of 2.2 times rated voltage applied between terminals for 3 ±2 seconds.

Barometric pressure:
Method 105 of MIL-STD-202, condition D (100, 000 ft).
Test potential: 100 percent of dc rated voltage.

Insulation resistance:
Method 302 of MIL-STD-202, condition A, 100 volts dc applied - 10, 000 megohms, minimum.

Capacitance:
Method 305 of MIL-STD-202.

DF: At 1 MHz ±100 kHz, at maximum and minimum capacitance - Shall be not more than 0.2 percent.

TABLE I. Style CV14.

Type designation	Capacitance		DC rated voltage	Characteristic				
				Symbol	Capacitance change from value at 25° C			
	(pf)		(volts)		At -55° C		At +85° C	
	Minimum	Maximum			Minimum Percent	Maximum Percent	Minimum Percent	Maximum Percent
CV14A070	1.5	7	500	A	-4.5	+2.0	-2.5	+2.0
CV14A120	3.0	12	500	A	-4.5	+2.0	-2.5	+2.0
CV14B130	3.0	13	500	B	-1.0	+3.5	-2.5	-0.5
CV14B200	5.0	20	500	B	-1.0	+3.5	-2.5	-0.5
CV14C300	4.0	30	500	C	-1.0	+6.5	-4.0	-1.0
CV14C450	7.0	45	500	C	-1.0	+6.5	-4.0	-1.0
CV14D300	4.0	30	500	D	+1.5	+7.0	-5.0	-1.5
CV14D450	7.0	45	500	D	+1.5	+7.0	-5.0	-1.5

Temperature coefficient:
Within the limits specified for the applicable characteristic.

Capacitance drift:
Within 0.75 percent of initial step 1 measurement or 0.50 picofarad, which ever is greater.

Terminal strength:
Pull test - Capacitor held by body and 4 pound load applied to each of terminals for at least 10 seconds.

Torque:
Not less than 4 nor more than 24 ounce-inches.

Fatigue:
ΔC - Shall not exceed 12% or 0.75 pf, whichever is greater.
Torque - Not less than 4 nor more than 36 ounce-inches.

Life:
Qualification test - 1, 000 hours at +85° C, 750 volts dc with a 100 Hertz or less 250 volt ac peak potential applied.
Insulation resistance - Initial requirement.
Capacitance change - Shall not exceed ±5 percent of initial value or 0.5 pf, whichever is greater.

Group C life: Conditions and requirements are the same as that required for qualification.

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Ⓐ Shock (specified pulse):
Method 213 of MIL-STD-202,
condition I (100 G).

Vibration:
Method 204 of MIL-STD-202, condition B (15G).
Capacitance change - Shall not exceed ±2 percent or 0.5 pf, whichever is greater.
DF - Shall be not more than **0.2 percent.**
Dielectric withstanding voltage:
1100 volts, dc applied for 3 ±2 seconds.
Insulation resistance:
10,000 megohms, minimum.

Moisture resistance:
Method 106 of MIL-STD-202:
Insulation resistance - 10,000 meg ms, minimum.
Capacitance change - Shall not exceed ±10 percent of initial value or **0.5 pf, whichever** is greater.
DF - Shall not be more than **0.5 percent**

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy - EC
Air Force - 11
DSA - ES

User activities:
Army -
Navy -
Air Force -

Preparing activity:
Air Force - 11

Agent:
DSA - ES

(Project 5910-1021-2)

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